1(a)	rate of change of velocity	[1]
1(b)	Gain in KE = Loss in GPE $\frac{1}{2}m(v_f^2 - v_i^2) = mg(h_i - h_f)$ $v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 187.846$ $v = 13.711 = 13.7$ or 14 m s^{-1} or by resolving components Vertically: $\uparrow: v^2 = u^2 + 2as$ $v^2 = (5.9 \sin 60^\circ)^2 + 2(-9.81)(1.2 - 9.0)$ $v_v = 13.3844 = 13.4$ Horizontally: $v_h = 5.9 \cos 60^\circ$ $v_h = 2.95$ Resultant velocity, $v = \sqrt{(13.4^2 + 2.95^2)} = 13.72 = 13.7$ or 14 m s^{-1}	[1] [1] [1] [1] [1] [1] [1]
1(ci)	W = mg = 78 × 9.81 = 765.18 = 765 N Resultant Force = 740 + 950 - 765 = 925 N Acceleration = F/m = 925/78 = 11.9 m s^{-2}	[1] [1] [1]
1(cii)	(As the diver moves down their) speed decreases (So) viscous force / drag (force) decreases	[1] [1]